FreshWay Gateway Service

Technical Installation Guide — Version 1.1

Mettler-Toledo FreshWay Scales

Applies to: Receipt, Label and Linerless printer configurations

Revision 1.1 (October 2026) — incorporates the transparent URL forwarding fix (RawUrl), the JposException response filter, and a new mandatory operating-system prerequisite: the HTTP.sys AllowRestrictedChars registry setting required for p_printnormal commands containing ESC sequences.

1. Architecture

The FreshWay printing stack consists of three Java processes managed by WinSW (Windows Service Wrapper), located in `D:\sdppc\RestService\Mettler Driver gateway\`.

The FreshWayGateway (C#) replaces only the gateway.jar (Spring Boot). The two MettlerWS backends continue running because they handle the hardware communication via JavaPOS → PAL-Service → USB. Replacing them is not necessary and not possible without a working LIBUSB installation.

```
Cache COS (%zprd11)
   |  HTTP :8082 (same port, same URLs - zero COS changes)
   v
FreshWayGateway.exe  [C# - replaces gateway.jar]
   +--> MettlerWSV1.1_G.jar :8083  (Label printer)
   +--> MettlerWSV1.1_B.jar :8084  (Receipt / Linerless printer)
            |  JavaPOS -> PAL-Service -> USB
            v
   FreshWay Printer Hardware
```

Expected improvement: gateway startup reduced from ~18 seconds (Spring Boot JVM) to under 1 second. Per-request latency reduced from ~50–100 ms to ~2–5 ms.

FreshWayPrinterService.cs (OPOS direct) was evaluated and discarded. It requires LIBUSB drivers that may not be correctly installed. FreshWayGateway.cs is the standard implementation.

v1.1 — transparent forwarding. The gateway forwards the request URL **exactly as received** (using the raw, undecoded URL). Cache URL-encodes special characters in print text (ESC as %1B, line feeds as %0A, escaped slashes as %01); the gateway must never decode and re-encode them, or the Java backend receives a corrupted path and answers 400 Bad Request.

2. Prerequisites

Verify on each server/scale before proceeding. If Java printing is working normally, all USB prerequisites are already met — the USB driver check is only needed after a fresh driver installation.

2.1 Java backends running

```
wmic process where "name='java.exe'" get ProcessId,CommandLine
```

Expected output:

```
java.exe -jar MettlerWSV1.1_G.jar 8083   (Label - Green)
java.exe -jar MettlerWSV1.1_B.jar 8084   (Receipt - Blue)
java.exe -Xmx256m -jar gateway.jar       (Gateway - to be replaced)
```

If only one backend appears, check which printer types are configured in Cache for this scale (MTA#1 = Receipt, MTA#2 = Label, MTA#3 = Linerless).

2.2 WinSW service IDs

Confirm the service IDs from the XML configuration files:

```
type "D:\sdppc\RestService\Mettler Driver gateway\WinSW.NET4.xml" | findstr "<id>"
type "D:\sdppc\RestService\Mettler Driver gateway\WinSW.G.xml"    | findstr "<id>"
type "D:\sdppc\RestService\Mettler Driver gateway\WinSW.B.xml"    | findstr "<id>"
```

Standard IDs (confirmed): `Driver_Gateway`, `Mettler_Driver_G`, `Mettler_Driver_B`.

2.3 .NET Framework

```
reg query "HKLM\SOFTWARE\Microsoft\NET Framework Setup\NDP\v4\Full" /v Release
```

Release value must be >= 460798 (4.7). Value 0x70bf6 = 4.7.2 is confirmed working.

2.4 HTTP.sys URL reservation

Run as Administrator — do this once per port per server:

```
REM Replace 8082 with the actual gateway port
netsh http delete urlacl url=http://+:8082/
netsh http add urlacl url=http://+:8082/ user=EVERYONE

REM Verify:
netsh http show urlacl | findstr "8082"
```

2.5 HTTP.sys restricted characters (MANDATORY — new in v1.1)

Background. The C# gateway uses HttpListener, which relies on the Windows kernel driver http.sys to parse incoming requests. By default, http.sys rejects with **400 Bad Request** any URL containing escaped characters that decode to U+0000–U+001F. Cache sends exactly such characters in `p_printnormal`: ESC (`%1B`) for FreshWay format commands (e.g. `ESC|1050uF` font sizing), line feeds (`%0A`) and escaped slashes (`%01`). Without this setting, those requests are rejected by the kernel **before they reach the gateway** and printing fails with error 400. The original gateway.jar was not affected because Spring Boot does not use http.sys.

Apply as Administrator (once per server), then reboot:

```
reg add "HKLM\SYSTEM\CurrentControlSet\Services\HTTP\Parameters" ^
  /v AllowRestrictedChars /t REG_DWORD /d 1

shutdown /r /t 0
```

The setting is read by http.sys at boot — a full restart is required. Verify after reboot:

```
reg query "HKLM\SYSTEM\CurrentControlSet\Services\HTTP\Parameters" ^
  /v AllowRestrictedChars
```

Expected: `AllowRestrictedChars REG_DWORD 0x1`

3. Compilation (one time)

Done once on any machine with .NET Framework 4.7+. The resulting FreshWayGateway.exe is copied to all scales.

3.1 Create working folder

```
mkdir C:\FreshwayPrinterService
cd C:\FreshwayPrinterService

REM Copy FreshWayGateway.cs and gateway.ini to this folder
```

3.2 Compile

```
"C:\Windows\Microsoft.NET\Framework64\v4.0.30319\csc.exe" ^
  /target:exe /platform:x64 ^
  /r:System.Web.dll /r:System.ServiceProcess.dll ^
  /out:FreshWayGateway.exe FreshWayGateway.cs
```

Expected output (no errors):

```
Microsoft (R) Visual C# Compiler version 4.7.3062.0 for C# 5
Copyright (C) Microsoft Corporation. All rights reserved.
```

Important (v1.1): always deploy the binary compiled from the current v1.1 source. Editing the .cs file does not change the running service — after any source change the service must be recompiled, the exe replaced, and the service restarted. Verify the exe timestamp when in doubt.

3.3 Create gateway.ini

```
; FreshWay Gateway Configuration
port=8082          ; Must match the port Cache uses for this printer
port_green=8083    ; MettlerWSV1.1_G.jar (Label printer)
port_blue=8084     ; MettlerWSV1.1_B.jar (Receipt / Linerless printer)
backend_timeout=30000
```

4. Installation per Scale

Repeat this section on each scale.

4.1 Copy files

Copy to `C:\FreshwayPrinterService\` on the target server:

- `FreshWayGateway.exe`
- `gateway.ini`

4.2 Edit gateway.ini

Confirm the port numbers match this scale's Java configuration:

```
REM Check which ports the Java backends use:
wmic process where "name='java.exe'" get CommandLine

REM Then edit:
notepad gateway.ini
```

4.3 Stop the Java gateway (keep backends running)

```
REM Stop ONLY the gateway - backends G and B must stay running
sc stop Driver_Gateway
sc config Driver_Gateway start= disabled

REM Verify backends are still running:
sc query Mettler_Driver_G | findstr STATE
sc query Mettler_Driver_B | findstr STATE
```

The backends (Green and Blue) must remain running. Only the gateway.jar process is being replaced.

4.4 Apply the HTTP.sys registry setting

If not done yet on this server, apply prerequisite 2.5 now (`AllowRestrictedChars=1`). The required reboot can be combined with the verification reboot of section 4.6.

4.5 Test in console mode

Open CMD as Administrator:

```
cd C:\FreshwayPrinterService
FreshWayGateway.exe --console
```

Expected output:

```
FreshWay Gateway v1.1
Escuchando en: http://+:8082/
Green (Label): 8083
Blue (Receipt): 8084
```

In another CMD window, test connectivity:

```
REM Receipt printer:
powershell -command "(New-Object System.Net.WebClient).DownloadString(
  'http://localhost:8082/pos/p_open/ReceiptPrinter-1')"
REM Expected: OK

REM Label printer (if configured on this scale):
powershell -command "(New-Object System.Net.WebClient).DownloadString(
  'http://localhost:8082/pos/p_open/LabelPrinter-1')"
REM Expected: OK

REM ESC sequence regression test (requires AllowRestrictedChars=1 + reboot):
powershell -command "(New-Object System.Net.WebClient).DownloadString(
  'http://localhost:8082/pos/p_printnormal/2/%1B%7C1050uF')"
REM Expected: request appears in the gateway console log (no 400)
```

Then trigger a test print from Cache and verify the gateway console shows the call sequence:

```
[ROUTE] ReceiptPrinter-1 -> :8084
[10:23:01.123] :8084  p_open        | ReceiptPrinter-1 | OK
[10:23:01.145] :8084  p_claim       | 10               | OK
[10:23:01.148] :8084  p_enable      | true             | OK
...
[10:23:01.910] :8084  p_close       |                  | OK
```

4.6 Install as Windows Service

```
sc create FreshWayGateway ^
  binPath= "C:\FreshwayPrinterService\FreshWayGateway.exe" ^
  DisplayName= "FreshWay Gateway Service" ^
  start= auto

sc description FreshWayGateway "C# HTTP gateway replacing gateway.jar for Cache COS"

sc failure FreshWayGateway reset= 60 actions= restart/5000/restart/10000/restart/30000

sc start FreshWayGateway
sc query FreshWayGateway
REM Must show:  STATE: 4  RUNNING
```

4.7 Verify after reboot

```
REM Registry setting must be present:
reg query "HKLM\SYSTEM\CurrentControlSet\Services\HTTP\Parameters" ^
  /v AllowRestrictedChars          <- 0x1

REM Java gateway must be disabled:
sc query Driver_Gateway | findstr STATE          <- STOPPED

REM Java backends must be running (if printers present):
sc query Mettler_Driver_G | findstr STATE        <- RUNNING
sc query Mettler_Driver_B | findstr STATE        <- RUNNING

REM C# gateway must be running:
sc query FreshWayGateway | findstr STATE         <- RUNNING

REM Port must be active:
netstat -ano | findstr "LISTEN" | findstr "8082"
```

Finish with a real receipt printed from Cache and check `^posfwlog`: no 400 errors must appear.

5. Configuration by Printer Type

The gateway routes based on the device name received in p_open. The routing table in FreshWayGateway.cs maps:

Device name (p_open)	Backend port	Backend
LabelPrinter-1	8083	MettlerWSV1.1_G.jar (Green)
LabelPrinter-1-Sim	8083	MettlerWSV1.1_G.jar (Green, simulator)
ReceiptPrinter-1	8084	MettlerWSV1.1_B.jar (Blue)
LinerlessPrinter-1	8084	MettlerWSV1.1_B.jar (Blue)

Scenarios. When a printer type is not present on a scale, its WinSW service should be disabled and both port_green / port_blue in gateway.ini pointed to the active backend port. On Receipt-only scales, Mettler_Driver_G is disabled; requests for Label devices then fail fast with a connection error, which is expected.

6. Enabling Label or Linerless on an Existing Scale

If a scale currently only has Receipt printing and needs to add Label or Linerless:

Hardware

- Verify the physical printer module is present in the scale.
- Connect and power on if needed.

Driver

- The UPOS 1.14 FLP package must be installed (MT-UPOS-1_14-POSPrinter-FLP MSI).
- Verify registration: `reg query "HKLM\SOFTWARE\OLEforRetail\ServiceOPOS\POSPrinter" /s` — "LabelPrinter-1 must appear in the registry output.
- The jpos.xml in the FLP installation directory must contain a LabelPrinter-1 entry. For Linerless, the FP backend's jpos.xml must contain a LinerlessPrinter-1 entry.

USB driver (only needed after driver reinstallation — e.g. if the Mettler DevconRemove BAT was run manually without the MSI installer; in normal installations USB drivers are set up automatically):

```
REM Power off the scale completely (not reboot) for 30 seconds, then power on

REM Check if the FLP device is visible:
devcon.exe findall "@USB\VID_0EB8&PID_300C*"

REM If found, install the WinUSB driver:
devcon.exe install LabelPrinterFLP2.inf  "USB\VID_0EB8&PID_300C"
devcon.exe install LabelPrinterFLP01.inf "USB\VID_0EB8&PID_3014"
```

Gateway configuration

- Enable the Green backend service: `sc config Mettler_Driver_G start= auto && sc start Mettler_Driver_G`
- Update gateway.ini: set `port_green=8083`
- Restart FreshWayGateway: `sc stop FreshWayGateway && sc start FreshWayGateway`
- Configure Cache for MTA#2 (Label) or MTA#3 (Linerless) pointing to the same gateway port.

7. Service Management

```
REM Start
sc start FreshWayGateway

REM Stop
sc stop FreshWayGateway

REM Restart
sc stop FreshWayGateway && sc start FreshWayGateway

REM Status
sc query FreshWayGateway

REM Uninstall
sc stop FreshWayGateway
sc delete FreshWayGateway
```

8. Troubleshooting

Symptom	Cause / Action
400 Bad Request on p_printnormal; request does NOT appear in the gateway console log	http.sys rejected the URL before it reached the gateway (escaped control characters: %1B, %0A, %01). Confirm in C:\Windows\System32\LogFiles\HTTPERR\httperr*.log (entries ending in '400 - URL'). Apply AllowRestrictedChars=1 (section 2.5) and reboot.
400 Bad Request on p_printnormal; request DOES appear in the console log	Running an outdated exe (pre-v1.1) that decodes the URL before forwarding. Recompile from the v1.1 source, replace the exe, restart the service. Check the exe timestamp.
Gateway listens but all requests fail with connection error	Backend not running or wrong port in gateway.ini. Verify with wmic / sc query (section 2.1) and match port_green / port_blue.
Label / Linerless requests fail, Receipt works	Expected on Receipt-only scales: Mettler_Driver_G is disabled. To enable Label, follow section 6.
^posfwlog shows java.lang.reflect.InvocationTargetException stack traces	Fixed in v1.1 (responses are filtered before returning to Cache). If seen, the running exe is outdated. A residual 'selectPrinter: not supported yet' JposException from directio cmd=100 is normal and ignored by Cache.
Nothing prints, gateway console shows OK sequence	Hardware/driver layer. Check the UPOS driver log D:\sdp_mta_prn_SCA01.txt and palservice.exe.

Useful live diagnostics: stop the service and run `FreshWayGateway.exe --console` to watch requests in real time; inspect Cache with `zw ^posfwlog` (clear with `k ^posfwlog`).

9. Cache COS Changes

No changes are required in the COS code.

The gateway exposes exactly the same HTTP endpoints on the same port (8082). The callWS function in %zprd11 continues to work unchanged. Cache does not know the backend changed from Java to C#.

The only configuration that may need adjustment is the port number in the Cache printer setup, if the new gateway uses a different port than the original gateway.jar. This is configured in the printer setup screen for each MTA#.

Appendix A — v1.1 Changelog

Change	Detail
Transparent URL forwarding	The gateway forwards ctx.Request.RawUrl instead of Url.AbsolutePath. AbsolutePath auto-decodes %XX sequences, which corrupted print text (spaces, line feeds) and caused the Java backend to answer 400. RawUrl preserves the exact bytes Cache sent.
JposException response filter	Backend responses starting with java.lang.reflect.InvocationTargetException are converted to an empty string for Cache, matching the behavior of the original Java gateway and keeping ^posfwlog clean.
HTTP.sys AllowRestrictedChars	New mandatory OS prerequisite (section 2.5): allows ESC/LF/SOH escaped characters in request URLs, required by p_printnormal format commands.
Known limitation / future direction	HttpListener depends on http.sys kernel parsing. A raw-socket (TcpListener) proxy variant is being evaluated to remove this dependency entirely. Not required for current deployments once AllowRestrictedChars is set.